

Tzu-Ching Yen

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WORK EXPERIENCE

Software Engineer at **Winterlight Labs** Toronto, Canada | 2021 - 2022

Worked on infrastructure that extracts acoustic and linguistic features in speech to monitor severity of neurodegenerative diseases

Research Assistant at **University of Toronto** Toronto, Canada | 2018 - 2022

- Proposed and investigated multiple quantum computing projects in Python and MATLAB that resulted in 15 publications and a patent collectively cited over 1000 times
- Presented at 8 conferences with audiences ranging from experts to undergraduate students
- Co-supervised four undergraduate research assistants, one master candidate, one PhD candidate, and one postdoctoral fellow in quantum computing projects

EDUCATION

2024 - present	PhD Candidate in Operation Research at Columbia University	
2022 - 2024	Masters's in Machine Learning at Carnegie Mellon University	(GPA: 4.17/4.30)
2017 - 2021	HBSC. with High Distinction at University of Toronto	(GPA: 3.96/4.00)
2014 - 2017	High School Diploma at Kent School	(GPA: 5.68/6.00)

PUBLICATIONS & PATENT

Published

1. Yen, T.-C. & Izmaylov, A. F. Cartan subalgebra approach to efficient measurements of quantum observables. *PRX Quantum* **2**, 040320 (2021).
2. Yen, T.-C., Verteletskyi, V. & Izmaylov, A. F. Measuring All Compatible Operators in One Series of Single-Qubit Measurements Using Unitary Transformations. *Journal of Chemical Theory and Computation* **16**. PMID: 32150412, 2400–2409 (2020).
3. Izmaylov, A. F., Yen, T.-C., Lang, R. A. & Verteletskyi, V. Unitary Partitioning Approach to the Measurement Problem in the Variational Quantum Eigensolver Method. *Journal of Chemical Theory and Computation* **16**. PMID: 31747266, 190–195 (2020).
4. Verteletskyi, V., Yen, T.-C. & Izmaylov, A. F. Measurement optimization in the variational quantum eigensolver using a minimum clique cover. *Journal of Chemical Physics* **152**, 124114 (2020).
5. Yen, T.-C., Lang, R. A. & Izmaylov, A. F. Exact and approximate symmetry projectors for the electronic structure problem on a quantum computer. *Journal of Chemical Physics* **151**, 164111 (2019).
6. Izmaylov, A. F., Yen, T.-C. & Ryabinkin, I. G. Revising the measurement process in the variational quantum eigensolver: is it possible to reduce the number of separately measured operators? *Chemical Science* **10**, 3746–3755 (2019).

7. Ryabinkin, I. G., Yen, T.-C., Genin, S. N. & Izmaylov, A. F. Qubit Coupled Cluster Method: A Systematic Approach to Quantum Chemistry on a Quantum Computer. *Journal of Chemical Theory and Computation* **14**. PMID: 30427679, 6317–6326 (2018).
8. Izmaylov, A. F. & Yen, T.-C. How to define quantum mean-field solvable Hamiltonians using Lie algebras. *Quantum Science and Technology* **6**, 044006 (2021).
9. Kottmann, J. S., Alperin-Lea, S., Tamayo-Mendoza, T., Cervera-Lierta, A., Lavigne, C., Yen, T.-C., Verteletskyi, V., Schleich, P., Anand, A., Degroote, M., Chaney, S., Kesibi, M., Curnow, N. G., Solo, B., Tsilimigkounakis, G., Zendejas-Morales, C., Izmaylov, A. F. & Aspuru-Guzik, A. TEQUILA: a platform for rapid development of quantum algorithms. *Quantum Science and Technology* **6**, 024009 (2021).
10. Izmaylov, A. F., Lang, R. A. & Yen, T.-C. Analytic gradients in variational quantum algorithms: Algebraic extensions of the parameter-shift rule to general unitary transformations. *Phys. Rev. A* **104**, 062443 (2021).
11. Yen, T.-C.*, Ganeshram*, A. & Izmaylov, A. F. Deterministic improvements of quantum measurements with grouping of compatible operators, non-local transformations, and covariance estimates. *npj Quantum Information* **9** (2023).
12. Choi, S., Yen, T.-C. & Izmaylov, A. F. Improving Quantum Measurements by Introducing “Ghost” Pauli Products. *Journal of Chemical Theory and Computation* **18**, 7394–7402 (2022).
13. Bansingh, Z. P., Yen, T.-C., Johnson, P. D. & Izmaylov, A. F. Fidelity Overhead for Nonlocal Measurements in Variational Quantum Algorithms. *The Journal of Physical Chemistry A* **126**, 7007–7012 (2022).
14. Martínez-Martínez, L. A., Yen, T.-C. & Izmaylov, A. F. Assessment of various Hamiltonian partitionings for the electronic structure problem on a quantum computer using the Trotter approximation. *Quantum* **7**, 1086 (2023).
15. Baby*, D., Garg*, S., Yen, T.-C.*, Balakrishnan, S., Lipton, Z. C. & Wang, Y.-X. Online Label Shift: Optimal Dynamic Regret meets Practical Algorithms. *Spotlight at Advances in Neural Information Processing Systems (NeurIPS)* (2023).
16. Patel, S., Yen, T.-C. & Izmaylov, A. F. Extension of Exactly-Solvable Hamiltonians Using Symmetries of Lie Algebras. *The Journal of Physical Chemistry A* **128**. PMID: 38718293, 4150–4159 (2024).
17. Patel, S., Jayakumar, P., Yen, T.-C. & Izmaylov, A. F. Quantum Measurement for Quantum Chemistry on a Quantum Computer. *Chemical Reviews* **ASAP**. PMID: 40690271 (2025).
18. Yen, T.-C., Siah, A. W. T., Chen, H., Peng, T., Guetta, D. & Namkoong, H. Data Mixture Optimization: A Multi-fidelity Multi-scale Bayesian Framework. *Advances in Neural Information Processing Systems (NeurIPS)* (2025).
19. Mittal, D., Li, A., Yen, T.-C., Guetta, D. & Namkoong, H. Architectural and Inferential Inductive Biases for Exchangeable Sequence Modeling. *Advances in Neural Information Processing Systems (NeurIPS)* (2025).
20. Yen, T.-C., Poeltl, J., Gear, H. S., Meng, Y., Fan, J., Shen, A., Liu, Y., Bauyrzhan, A., Shea, P., Du, S., Liu, H., Guetta, D. & Namkoong, H. MBABench: Evaluating LLM Agents on End-to-End Spreadsheet Tasks in Finance. *Advances in Neural Information Processing Systems (NeurIPS), Evaluations and Datasets Track* (2026).

Submitted

21. Mittal*, D., Castellani*, T., Yen, T.-C.*, Ye, N., Wu, F., Chen, M., Cai, T., Koukoumidis, E., Zeng, W. & Namkoong, H. LatentGym: A Testbed For Cross-Task Experiential Learning With Controllable Latent Structure, arXiv:2606.15306 (2026).
22. Yang, G., Chen, Y., Yen, T.-C. & Namkoong, H. Benchmarking In-context Experiential Learning Through Repeated Product Recommendations, arXiv:2511.22130 (2025).
23. Castellani, T., Ye, N., Mittal, D., Yen, T.-C., Koukoumidis, E., Zeng, W. & Namkoong, H. Synth-Tools: A Framework for Scaling Synthetic Tools for Agent Development, arXiv:2511.09572 (2025).

Patents

24. Izmaylov, A. F., Anand, A., Kottmann, J. S., Aspuru-Guzik, A., Lang, R. A. & Yen, T.-C.. Operator Implementations for Quantum Computation, United States, 63/222, 546, 2021/07/16.